

$$1. \int_0^{\pi/2} q \cos^2(\theta) d\theta = q \int_0^{\pi/2} \cos^2(\theta) d\theta$$

Half angle $\cos^2(\theta) = \frac{1}{2}(1 + \cos(2\theta)) = \frac{1}{2} + \frac{1}{2}\cos(2\theta)$

$$q \int_0^{\pi/2} \left(\frac{1}{2} + \frac{1}{2}\cos(2\theta) \right) d\theta = q \left[\int_0^{\pi/2} \frac{1}{2} d\theta + \int_0^{\pi/2} \cos(2\theta) d\theta \right]$$

$$\int \frac{1}{2} d\theta = \frac{1}{2} \theta \quad \int \cos(2\theta) d\theta \quad u=2\theta \quad \frac{du}{d\theta}=2 \quad du=2d\theta \quad \theta=\frac{1}{2}du$$

$$\int_0^{\pi/2} \cos(u) \frac{1}{2} du = \frac{1}{2} \int_0^{\pi/2} \cos(u) du = \frac{1}{2} \sin(u) = \frac{1}{2} \sin(2\theta)$$

$$= q \left[\frac{1}{2} \theta + \frac{1}{2} \left[\frac{1}{2} \sin(2\theta) \right] \right] \Big|_0^{\pi/2}$$

$$q \left[\left(\frac{1}{2} \cdot \frac{\pi}{2} \right) + \frac{1}{2} \left[\frac{1}{2} \sin\left(\frac{2\pi}{2}\right) \right] \right] = q \left[\left(\frac{\pi}{4} \right) + \frac{1}{2} \left[\frac{1}{2} \sin(\pi) \right] \right]$$

$$= q \left[\frac{\pi}{4} + \frac{1}{2} \left(\frac{\pi}{2} \cdot 0 \right) \right] = q \left[\frac{\pi}{4} \right] = \frac{q\pi}{4}$$

$$q \left[\left(\frac{1}{2} \cdot 0 \right) + \frac{1}{2} \left[\frac{1}{2} \sin(0) \right] \right] = q \left[(0) + \frac{1}{2} (0) \right] = 0$$

$$\frac{q\pi}{4} - 0 = \frac{q\pi}{4}$$

2 $\tan^2(x) \sec(x) dx$

Already gone over in notes

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$$\int \cos^2(x) \tan^3(x) dx = \int \cos^2(x) \cdot \frac{\sin^3(x)}{\cos^3(x)} dx = \int \frac{\cos^2(x) \cdot \sin^3(x)}{\cos^3(x)} dx = \int \frac{\sin^3(x)}{\cos(x)} dx = \int \sin^2(x) \cdot \frac{\sin(x)}{\cos(x)} dx$$

Half angle identity - $\sin^2(\theta) + \cos^2(\theta) = 1$

$$\tan^3(x) = \left(\frac{\sin(x)}{\cos(x)}\right)^3 = \frac{\sin^3(x)}{\cos^3(x)}$$

$$\Rightarrow \int \sin^2(x) \cdot \frac{\sin(x)}{\cos(x)} dx \quad \text{Let } \sin^2(x) = 1 - \cos^2(x)$$

$$\Rightarrow \int \sin(x) \cdot \frac{1 - \cos^2(x)}{\cos(x)} dx \quad u = \cos(x) \quad \frac{du}{dx} = -\sin(x) \Rightarrow du = -\sin(x) dx$$

$$\text{Let } \int \sin(x) \cdot \frac{1 - u^2}{u} du = -\int \frac{1 - u^2}{u} du \quad (-1) du = \sin(x) dx$$

$$= -\int \frac{1}{u} du + \int u du = -\int \frac{1}{u} du + \frac{1}{2} u^2$$

$$= -\int \frac{1}{u} du + \frac{1}{2} u^2 + C = -\int \frac{1}{\cos(x)} dx + \frac{1}{2} \cos^2(x) + C$$

$$\int \cos^2(x) \cdot \tan^3(x) dx$$

$$= \int \cos^2(x) \cdot \frac{\sin^3(x)}{\cos^3(x)} dx = \int \cos^2(x) \cdot \sin(x) \cdot \frac{\sin^2(x)}{\cos^3(x)} dx \quad \tan^3(x) = \left(\frac{\sin(x)}{\cos(x)}\right)^3 = \frac{\sin^3(x)}{\cos^3(x)}$$

Half angle identity - $\sin^2(\theta) + \cos^2(\theta) = 1$

$$\Rightarrow \sin^2(x) = 1 - \cos^2(x)$$

$$= \int \cos^2(x) \cdot \sin(x) \cdot \frac{1 - \cos^2(x)}{\cos^3(x)} dx = \int \cos^2(x) \cdot \sin(x) \cdot \frac{(1 - \cos^2(x)) \cos^2(x)}{\cos^3(x)} dx$$

$$= \int \sin(x) \cdot \frac{1 - \cos^2(x)}{\cos(x)} dx \quad u = \cos(x) \quad \frac{du}{dx} = -\sin(x)$$

$$= \int -1 \cdot \frac{1 - u^2}{u} du = -\int \frac{1 - u^2}{u} du = -\int \frac{1}{u} du + \int u du =$$

$$-\left[\int \frac{1}{u} du - \int u du\right] = -\left[\ln|u| - \left(\frac{1}{2} u^2\right)\right]$$

$$= -\left[\ln|\cos(x)| - \left(\frac{1}{2} \cos^2(x)\right) + C\right]$$

$$4 \quad 55 (1 + \cos(\theta))^2 d\theta = 55 (1 + \cos(\theta))^2 d\theta$$

$$(1 + \cos \theta) \left(\frac{1}{2} + \cos \theta \right)$$

$$1 + \cos \theta + \cos \theta + \cos^2 \theta = (1 + 2\cos \theta) + \cos^2 \theta$$

$$55 (1 + 2\cos \theta) + \cos^2 \theta d\theta$$

$$\text{Hilfanglei: } \cos^2 \theta = \frac{1}{2} (1 + \cos(2\theta))$$

$$55 \left(1 + 2\cos \theta + \frac{1}{2} (1 + \cos(2\theta)) \right) d\theta = 55 \left(1 + 2\cos \theta + \frac{1}{2} + \frac{1}{2} \cos(2\theta) \right) d\theta$$

$$55 \left(\frac{3}{2} + 2\cos \theta + \frac{1}{2} \cos(2\theta) \right) d\theta$$

$$= 5 \left[\int \frac{3}{2} d\theta + \int 2 \cos \theta d\theta + \int \frac{1}{2} \cos(2\theta) d\theta \right]$$

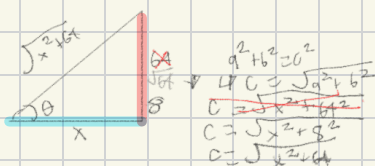
$$\int \frac{3}{2} d\theta = \frac{3}{2} \theta$$

$$\int 2 \cos \theta d\theta = 2 \sin \theta$$

$$\begin{aligned} & \int \frac{1}{2} \cos(2\theta) d\theta \quad u = 2\theta \\ & \quad \frac{du}{d\theta} = 2 \quad d\theta = \frac{1}{2} du \\ & = \frac{1}{2} \int \cos(u) \frac{1}{2} du \\ & = \frac{1}{2} \cdot \frac{1}{2} \int \cos u du \\ & = \frac{1}{4} [\sin u] = \frac{1}{4} \sin(2\theta) \end{aligned}$$

$$= 5 \left[\frac{3}{2} \theta + 2 \sin \theta + \frac{1}{4} \sin(2\theta) + C \right]$$

$$5 - \int \frac{x^3}{\sqrt{x^2 + 64}} dx$$

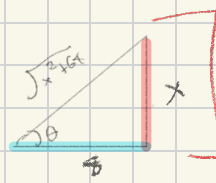


$$\text{but } x = 8 \tan \theta$$

$$\rightarrow \theta = \arctan \frac{x}{8}$$

So x would be $8 \cdot \frac{8}{x}$ if this triangle is true. And x should simplify to x .

Some correct triangles should be...



Solns is the correct triangle $\sin \theta = \frac{x}{\sqrt{x^2 + 64}}$

$$\text{So } x = 8 \tan \theta \rightarrow x = 8 \frac{x}{8} = \frac{8x}{8} \Rightarrow$$

Now to actually do the integral... for $\sqrt{x^2 + a^2}$ (and $a=8$) $x = 8 \tan \theta$

$$x = 8 \tan \theta$$

$$\int \frac{x^3}{\sqrt{x^2 + 64}} dx$$

$$x = 8 \tan \theta \rightarrow \frac{dx}{d\theta} = 8 \sec^2(\theta)$$

$$dx = 8 \sec^2(\theta) d\theta$$

$$\text{Num} = x^3$$

$$\text{den} = \sqrt{x^2 + 64}$$

$$x^3 = (8 \tan \theta)^3 = 8^3 \tan^3 \theta = 512 \tan^3 \theta$$

$$\text{den} = \sqrt{(8 \tan \theta)^2 + 64} = \sqrt{64 \tan^2 \theta + 64} = \sqrt{64(\tan^2 \theta + 1)} = \sqrt{64 \sec^2 \theta}$$

$$= 8 \sec \theta$$

$$\int \frac{x^3}{\sqrt{x^2 + 64}} dx = \int \frac{512 \tan^3 \theta}{8 \sec \theta} \cdot 8 \sec^2(\theta) d\theta$$

$$= \int \frac{512 \tan^3 \theta}{8 \sec \theta} \cdot 8 \sec^2(\theta) d\theta$$

$$d\theta = \int 512 \tan^3 \theta \sec \theta d\theta$$

$$= 512 \int \tan^3 \theta \sec \theta d\theta = 512 \int \tan^2 \theta \tan \theta \sec \theta d\theta$$

$$= 512 \int (\sec^2 \theta - 1) \tan \theta \sec \theta d\theta$$

$$= 512 \int (\sec^2 \theta - 1) d\theta = 512 \left[\frac{1}{3} \sec^3 \theta - \sec \theta \right] + C$$

$$= 512 \left[\frac{1}{3} \sec^3 \theta - \sec \theta \right] + C$$

$$= 512 \left[\frac{1}{3} \left(\frac{\sqrt{x^2 + 64}}{8} \right)^3 - \frac{\sqrt{x^2 + 64}}{8} \right] + C$$

$$\sec = \frac{1}{\cos \theta} = \frac{1}{\frac{8}{\sqrt{x^2 + 64}}} = \frac{\sqrt{x^2 + 64}}{8}$$

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$$\int \sqrt{x^2 + 14x} \, dx$$

$$x^2 + 14x = \left(x^2 + 14x + \left(\frac{14}{2}\right)^2\right) - \left(\frac{14}{2}\right)^2 =$$

$$(x+7)^2 - 7^2$$

$$= \int \sqrt{(x+7)^2 - 49} \, dx \quad v = x+7 \quad \frac{dv}{dx} = 1 \quad dv = dx$$

$$= \int \sqrt{v^2 - 49} \, dv \text{ then trig sub}$$

$$= \int \sqrt{v^2 - a^2} \, dv \text{ where } a = 7 \quad \frac{dv}{d\theta} = 7 \sec \theta \tan \theta$$

$$v = 7 \sec \theta \quad dv = 7 \sec \theta \tan \theta \, d\theta$$

$$= \int \sqrt{v^2 - 49} \, 7 \sec \theta \tan \theta \, d\theta$$

$$\int 7 \sec \theta \sqrt{7^2 \sec^2 \theta - 49} \, d\theta \quad v = 7 \sec \theta \quad \tan^2 \theta + 1 = \sec^2 \theta \quad \tan^2 \theta = \sec^2 \theta - 1$$

$$= \int 49 \sec^2 \theta - 49 = \int 49 (\sec^2 \theta - 1) = \int 49 (\tan^2 \theta) = \int 49 \tan^2 \theta \, d\theta$$

$$\int 7 \tan \theta \, 7 \sec \theta \tan \theta \, d\theta$$

$$= \int 7 \tan^2 \theta \, 7 \sec \theta \, d\theta$$

$$= \int 49 \tan^2 \theta \sec \theta \, d\theta$$

$$= 49 \int \tan^2 \theta \sec \theta \, d\theta = \frac{\sec(x) \tan(x) - \ln |\sec(x) + \tan(x)|}{2} + C$$

$$= 49 \frac{\sec \theta \tan \theta - \ln |\sec \theta + \tan \theta|}{2} + C$$

$$v = 7 \sec \theta \quad \sec \theta = \frac{1}{\cos \theta} = \frac{H}{A} \quad \text{Now sub } \theta \text{ back for } v \text{ then } v \text{ back for } x$$



$$v = 7 \cdot \frac{H}{A} = \frac{7H}{A} = \frac{7W}{1} = 7W$$

$$= 49 \frac{v \cdot \frac{W}{7} - \ln |v + \frac{W}{7}|}{2} + C$$

$$= 49 \left[\frac{1}{2} \left[\left(v \cdot \left(\frac{1}{7} \sqrt{v^2 - 49} \right) \right) - \ln \left| v + \left(\frac{1}{7} \sqrt{v^2 - 49} \right) \right| \right] \right] + C$$

$$= 49 \left[\frac{1}{2} \left[\left((x+7) \left(\frac{1}{7} \sqrt{(x+7)^2 - 49} \right) \right) - \ln \left| (x+7) + \left(\frac{1}{7} \sqrt{(x+7)^2 - 49} \right) \right| \right] \right] + C$$

Doublecheck

wrong

$$\int \sqrt{x^2 + 14x} \, dx$$

$$x^2 + 14x = (x^2 + 14x + (\frac{7}{2})^2) - (\frac{7}{2})^2 = (x + \frac{7}{2})^2 - (\frac{7}{2})^2$$

$$C = 14$$

$$= (x + \frac{14}{2})^2 - (\frac{14}{2})^2 = (x+7)^2 - (7)^2 = (x+7)^2 - 49$$

$$\int \sqrt{u^2 - 49} \, du = \int \sqrt{u^2 - 7^2} \, du$$



$$\begin{aligned} \sqrt{(x+7)^2 - 49} &= \sqrt{x^2 + 14x} \\ &= \sqrt{x^2 + 14x} \end{aligned}$$

$$\sin \theta = \frac{x+7}{u} \Rightarrow \frac{\sqrt{(x+7)^2 - 49}}{x+7} = \frac{\sqrt{u^2 - 49}}{u}$$

$$\cos \theta = \frac{7}{u} \quad \tan \theta = \frac{\sqrt{x^2 + 14x}}{7} = \frac{\sqrt{u^2 - 49}}{7}$$

$$\sec \theta = \frac{(x+7)}{7} = \frac{u}{7}$$

$$7 \sec \theta = u \quad \frac{du}{d\theta} = 7 \sec \theta \tan \theta$$

$$du = 7 \sec \theta \tan \theta \, d\theta$$

$$\int \sqrt{u^2 - 49} \, dx = \int 7 \tan \theta \, du = \int 7 \tan \theta \cdot (7 \sec \theta \tan \theta \, d\theta) = 49 \int \tan^2 \theta \sec \theta \, d\theta$$

$$= 49 \int \tan^2 \theta \sec \theta \, d\theta$$

$$= 49 \int (\sec^2 \theta - 1) \sec \theta \, d\theta$$

$$= 49 \int (\sec^3 \theta - \sec \theta) \, d\theta$$

$$= 49 \left[\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| - \ln |\sec \theta + \tan \theta| \right]$$

$$= 49 \left[\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| - \ln |\sec \theta + \tan \theta| \right]$$

$$= 49 \left[\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| - \ln |\sec \theta + \tan \theta| \right]$$

$$= 49 \left[\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| - \ln |\sec \theta + \tan \theta| \right]$$

$$= \frac{1}{2} (\ln |\sec \theta + \tan \theta|) - \frac{1}{2} (\ln |\sec \theta + \tan \theta|)$$

$$= -\frac{1}{2} \ln |\sec \theta + \tan \theta|$$

$$= 49 \left[\frac{1}{2} (\sec \theta \tan \theta) - \frac{1}{2} (\ln |\sec \theta + \tan \theta|) \right] = \frac{49}{2} (\sec \theta \tan \theta) - \frac{49}{2} (\ln |\sec \theta + \tan \theta|) + C$$

$$\sec \theta = \frac{u}{7}$$

$$\cos \theta = \frac{7}{u}$$



$$\tan \theta = \frac{x+7}{7}$$

$$\sqrt{x^2 + 14x}$$

$$\tan \theta = \frac{\sqrt{x^2 + 14x}}{7}$$

$$\sec \theta = \frac{x+7}{7}$$

$$\frac{49}{2} \left(\frac{x+7}{7} \cdot \frac{\sqrt{x^2 + 14x}}{7} \right) - \frac{49}{2} \left(\ln \left| \frac{x+7}{7} + \frac{\sqrt{x^2 + 14x}}{7} \right| \right) + C$$

$$E x \sqrt{x^2 + 2x} \quad dx$$

$$= \sqrt{x^2 + 2x} = \sqrt{(x+1)^2 - 1}$$

$$x+1 = \sec \theta$$

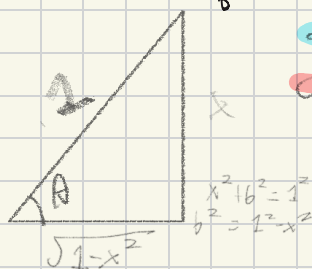
$$= \frac{\sqrt{\sec^2 \theta - 1}}{\tan \theta} = \frac{\tan^2 \theta}{\tan \theta} = \tan \theta$$

$$\int \sqrt{(x+1)^2 - 1} dx = \int \tan \theta \sec \theta \tan \theta d\theta$$

$$= \frac{1}{2} (x+1) \sqrt{x^2 + 2x} - \frac{1}{2} \ln(x+1 + \sqrt{x^2 + 2x}) + C$$

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$$\int_0^1 x^3 \sqrt{1-x^2} dx$$



$$\sin \theta = \frac{x}{1} = x \quad x = \sin \theta \quad \frac{dx}{d\theta} = \cos \theta$$

$$\cos \theta = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2} \quad dx = \cos \theta d\theta$$

$$\int x^3 \sqrt{1-x^2} dx = \int \sin^3 \theta \cdot \cos \theta dx$$

$$= \int \sin^3 \theta \cos \theta \cos \theta d\theta = \int \sin^3 \theta \cos^2 \theta d\theta$$

We have $\sin^m$ and $\cos^n$, $m=3$ so m is odd

$$\int \sin^2 \theta \sin \theta \cos^2 \theta d\theta \quad \text{Pythagoras: } \sin^2 \theta = 1 - \cos^2 \theta$$

$$= \int (1 - \cos^2 \theta) \sin \theta \cos^2 \theta d\theta$$

$$= \int (1 - u^2) \sin \theta u^2 d\theta$$

$$= \int (1 - u^2) u^2 du (-1)$$

$$= -1 \int (1 - u^2) u^2 du = -1 \int u^2 - u^4 du = -1 \left[\frac{u^3}{3} - \frac{u^5}{5} \right]$$

$$u = \cos \theta \quad \frac{du}{d\theta} = -\sin \theta$$

$$du = -\sin \theta d\theta$$

$$du \cdot (-1) = \sin \theta d\theta$$

$$= -1 \left[\frac{u^3}{3} - \frac{u^5}{5} \right]$$

$$= -\frac{u^3}{3} + \frac{u^5}{5} + C$$

$$4 \left[-\frac{u^3}{3} + \frac{u^5}{5} + C \right] \Big|_0^1$$

$$u = \cos \theta$$

$$4 \left[\left(-\frac{1}{3} \cos^3 \theta \right) + \left(\frac{1}{5} \cos^5 \theta \right) + C \right] \Big|_0^1$$

$$\cos \theta = \sqrt{1-x^2}$$

$$4 \left[\left(-\frac{1}{3} (\sqrt{1-x^2})^3 \right) + \left(\frac{1}{5} (\sqrt{1-x^2})^5 \right) \right] \Big|_0^1$$

$$4 \left[\left(-\frac{1}{3} (\sqrt{1-1})^3 \right) + \left(\frac{1}{5} (\sqrt{1-1})^5 \right) \right]$$

$$= 4 \left[\left(-\frac{1}{3} (0)^3 \right) + \left(\frac{1}{5} (0)^5 \right) \right] = 0$$

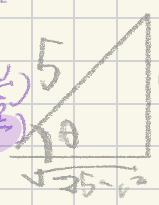
$$4 \left[\left(-\frac{1}{3} (\sqrt{1})^3 \right) + \left(\frac{1}{5} (\sqrt{1})^5 \right) \right] = \left(-\frac{4}{3} (\sqrt{1})^3 \right) + \left(\frac{4}{5} (\sqrt{1})^5 \right)$$

$$= -\frac{4\sqrt{1}^3}{3} + \frac{4\sqrt{1}^5}{5}$$

$$= - \left(-\frac{4\sqrt{1}^3}{3} + \frac{4\sqrt{1}^5}{5} \right)$$

$\int \sqrt{13 + 12x - x^2} dx$

$u = 5 \sin \theta$
 $\sin \theta = \frac{u}{5}$
 $\theta = \arcsin\left(\frac{u}{5}\right)$
 $= \arcsin\left(\frac{x - \sqrt{12}}{5}\right)$



$\sin \theta = \frac{u}{5}$
 $\cos \theta = \frac{\sqrt{25-u^2}}{5}$
 $u^2 + b^2 = 5^2$
 $u^2 = 25 - b^2$
 $\rightarrow 5 \cos \theta = \sqrt{25 - u^2}$

$u = 5 \sin \theta$
 $\sin \theta = \frac{u}{5}$

$\frac{du}{d\theta} = 5 \cos \theta \Rightarrow du = 5 \cos \theta d\theta$

$u = x - \sqrt{12} \Rightarrow x = u + \sqrt{12}$
 $x = 5 \sin \theta + \sqrt{12}$

$\frac{dx}{d\theta} = 5 \cos \theta \Rightarrow dx = 5 \cos \theta d\theta$

$\int \sqrt{13 + 12x - x^2} dx = \int \sqrt{25 - u^2} \cdot 5 \cos \theta d\theta$

$= \int 5 \cos \theta \cdot 5 \cos \theta d\theta = 25 \int \cos^2 \theta d\theta$

$= \frac{25}{2} \int (1 + \cos(2\theta)) d\theta$

$= \frac{25}{2} \left[\theta + \frac{1}{2} \sin(2\theta) \right] + C$

$= \frac{25}{2} \left[\theta + (\sin \theta \cos \theta) \right] + C$

$= \left(\frac{25}{2} \arcsin\left(\frac{x - \sqrt{12}}{5}\right) \right) + \frac{25}{2} \left(\frac{u}{5} \cdot \frac{\sqrt{25 - u^2}}{5} \right) + C$

$= \left(\frac{25}{2} \arcsin\left(\frac{x - \sqrt{12}}{5}\right) \right) + \frac{25}{2} \left(\frac{x - \sqrt{12}}{5} \cdot \frac{\sqrt{25 - (x - \sqrt{12})^2}}{5} \right) + C$

Expansion is incorrect

$$\int \sqrt{13 + 12x - x^2} dx$$

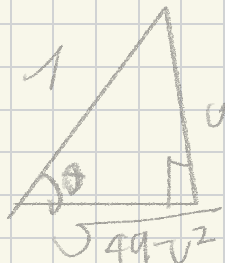
$$-x^2 + 12x + 13 = -(x^2 - 12x) + 13$$

$$= -(x^2 - 12x + 36 - 36) + 13$$

$$= -(x - 6)^2 - 36 + 13$$

$$= -(x - 6)^2 + 49 \quad v = x - 6$$

$$\int \sqrt{-(x-6)^2 + 49} = \int \sqrt{-v^2 + 49} = \int \sqrt{7^2 - v^2} \quad \text{where } a=7$$



$$7^2 = v^2 + b^2 = 7 \cos \theta dx$$

$$49 - v^2 = b^2$$

$$\sin \theta = \frac{v}{7}$$

$$\cos \theta = \frac{\sqrt{49-v^2}}{7} \quad v = 7 \sin \theta$$

$$\sqrt{49-v^2} = 7 \cos \theta$$

$$v = 7 \sin \theta \rightarrow \sin \theta = \frac{v}{7} \rightarrow \theta = \arcsin\left(\frac{v}{7}\right)$$

$$\frac{dv}{d\theta} = 7 \cos \theta \rightarrow dv = 7 \cos \theta d\theta = 5 \cdot 7 \cos \theta \cdot 7 \cos \theta d\theta$$

$$v = x - 6 \quad x = v + 6 = 49 \cos^2 \theta d\theta \quad 49 \int \cos^2 \theta d\theta$$

$$x = 7 \sin \theta + 6$$

$$\frac{dx}{d\theta} = 7 \cos \theta \quad dx = 7 \cos \theta d\theta = \frac{49}{2} [\sin \theta + \sin(2\theta)] d\theta$$

$$= \frac{49}{2} \left[\theta + \frac{1}{2} \sin(2\theta) \right] + C = \frac{49}{2} \left[\theta + \left(\frac{1}{2} (2 \sin \theta \cos \theta) \right) \right] + C$$

$$= \frac{49}{2} \left[\theta + (\sin \theta \cos \theta) \right] + C = \frac{49}{2} \theta + \frac{49}{2} \sin \theta \cos \theta + C$$

$$= \left(\frac{49}{2} \arcsin\left(\frac{v}{7}\right) \right) + \frac{49}{2} \left(\frac{\sqrt{49-v^2}}{7} \right) + C$$

$$= \left(\frac{49}{2} \arcsin\left(\frac{x-6}{7}\right) \right) + \frac{49}{2} \left(\frac{\sqrt{13+12x-x^2}}{7} \right) + C$$

9. $\tan^2(x) - \sec^2(x)$ - write in terms of cos
Then simplify

$$\tan^2 \Rightarrow \frac{\sin^2}{\cos^2}$$

$$\sec^2 = \frac{1}{\cos^2}$$

$$\frac{\sin^2}{\cos^2} - \frac{1}{\cos^2}$$

same denom so can subtract

$$\frac{\sin^2 - 1}{\cos^2} =$$

$$\sin^2 - 1 = -\cos^2$$

$$= \frac{-\cos^2}{\cos^2} = -1$$

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$$\int \frac{dx}{(x^2 - 1)^{3/2}}$$

$$u = dx \cdot \frac{1}{(x^2 - 1)^{3/2}} = (x^2 - 1)^{-3/2} dx = \int \frac{1}{(x^2 - 1)^{3/2}} dx$$

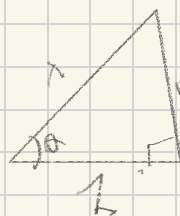
$$x = \sec \theta$$



$$1^2 + b^2 = x^2$$

$$b^2 = x^2 - 1^2$$

$$\sec \theta = \frac{1}{\cos} = \frac{x}{1} = \frac{x}{\sqrt{x^2 - 1}} \quad \text{Switch legs}$$



$$\sec \theta = \frac{H}{A} = \frac{x}{1} = x$$

$$\tan \theta = \frac{\sqrt{x^2 - 1}}{1} = \sqrt{x^2 - 1}$$

$$\cos \theta = \frac{1}{x}$$

$$1 = \cos \theta \cdot x$$

$$1 = \cos \theta \sec \theta$$

$$\frac{dx}{d\theta} = \sec \theta \tan \theta$$

$$dx = \sec \theta \tan \theta d\theta$$

$$= \int \frac{1}{\sec^3 \theta} dx = \int \frac{1}{\tan^3 \theta} \sec \theta \tan \theta d\theta$$

$$= \int \frac{\sec \theta \tan \theta}{\tan^3 \theta} d\theta = \int \frac{\sec \theta}{\tan^2 \theta} d\theta$$

$$= \int \frac{1}{\cos \theta} \cdot \frac{1}{\tan^2 \theta} d\theta = \int \frac{1}{\cos \theta} \cdot \frac{\cos^2 \theta}{\sin^2 \theta} d\theta = \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

$$v = \sin \theta \quad dv = \cos \theta d\theta \quad \int \frac{1}{v^2} dv = \int v^{-2} dv$$

$$= -\frac{1}{v} + C = -\frac{1}{\sin \theta} + C = -\frac{1}{\sin(\sec^{-1}(x))} + C$$

$$x = \sec \theta$$

$$\theta = \sec^{-1}(x)$$

$$= -\frac{1}{\sin(\sec^{-1}(x))} + C$$